

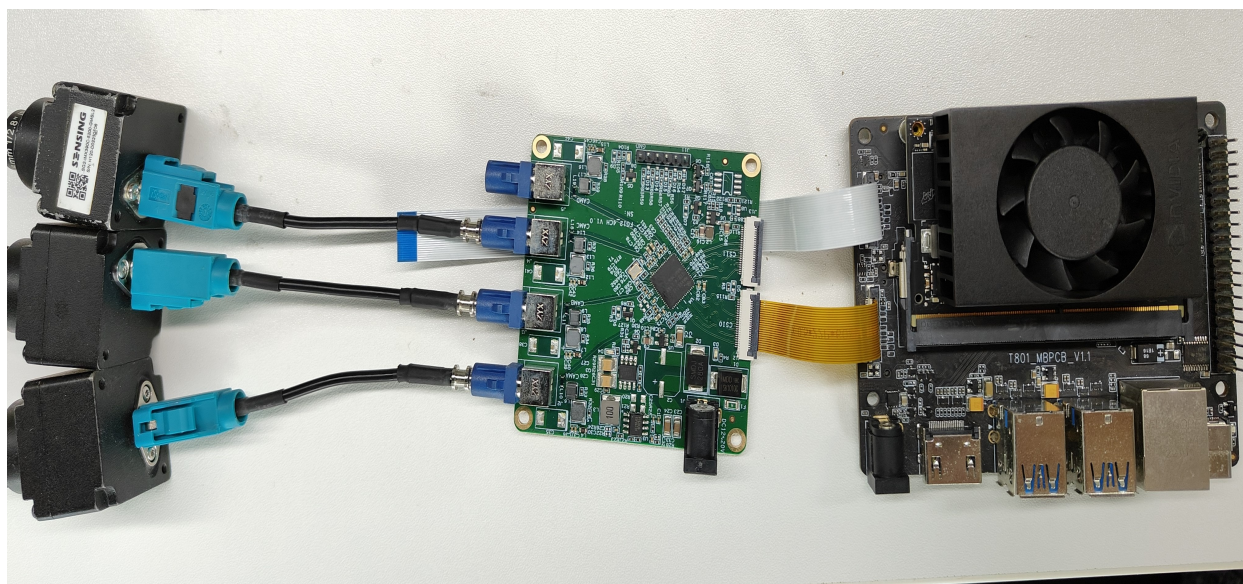


Product Introduction



- FG12_4CH GMSL Board is a 4-channel GMSL capture board developed by Zhuying Intelligent / Fangzhu Technology based on the ADI Deserializer chip MAX96712, hereinafter referred to as the Capture Board.
- This capture board can connect up to 4 GMSL cameras simultaneously and output CSI signals to the NVIDIA® Jetson Orin™ NX Development Kit (it can also be connected to the Xavier NX Development Kit).
- Through software configuration, the MAX96712 can adaptively decode GMSL cameras with different rates of GMSL1 and GMSL2 protocols.
- The 22Pin connector on the NVIDIA Jetson® Orin/Xavier™ NX Development Kit cannot supply the power required by the cameras. Therefore, the capture board is equipped with a pluggable external DC power connector for 12V~24V input (the same power adapter as the kit can be used).

Product Image



Document Version

Document Version	Revision Date	Revision Content
V1.0	2025/09/09	Initial Release
V1.1	2026/01/13	Content Revision

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Before using this manual, please read the following license agreement carefully. You may only use the products introduced in this manual if you agree to the terms of this license agreement.

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Technical Support and Information Feedback

If you encounter any problems when using our products, or if you find any functional defects in our products, you may visit our official website: www.joiningai.com to contact our customer service team for problem resolution and feedback. For technical support, guidance or any valuable suggestions, please also contact us via our official website or phone call.

Warranty Terms

Warranty Period: The warranty period for all equipment of Zhuying Intelligent Robot Co., Ltd. is 1 year from the date of purchase.

Warranty Clause: For products with non-human-caused malfunctions within the warranty period, Zhuying Intelligent Robot (Shenzhen) Co., Ltd. will provide free warranty services. Please obtain warranty assistance through customer service chat on the purchase platform or via phone call (for details, refer to the Warranty Terms of Zhuying Intelligent Robot (Shenzhen) Co., Ltd.).

Contact Information

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Safety Instructions

Before using this product, you must read this document first to gain a basic understanding of the product, and strictly comply with the safety instructions in this user manual to ensure your personal safety and prevent equipment damage. The manufacturer shall not be liable for any damage to equipment or loss of personal life and property caused by blind operation and improper use.

Personal Safety

Before operating the equipment, wear anti-static work clothes, anti-static gloves or wrist straps, and remove conductive objects such as jewelry and watches to avoid electric shock or thermal burns.

In the event of a fire, evacuate the building or equipment area immediately, activate the fire alarm bell or dial the local fire emergency number. Under no circumstances shall anyone re-enter a burning building.

Power Supply Voltage

FG12_4CH_GMSL carrier board supports an input power range of 12V ~ 24V with a current of 2A or above.

Environmental Requirements

Operating Temperature : -20°C - 65°C

Ventilation Requirements : Adequate ventilation must be ensured around the installed computing platform.

Grounding Requirements

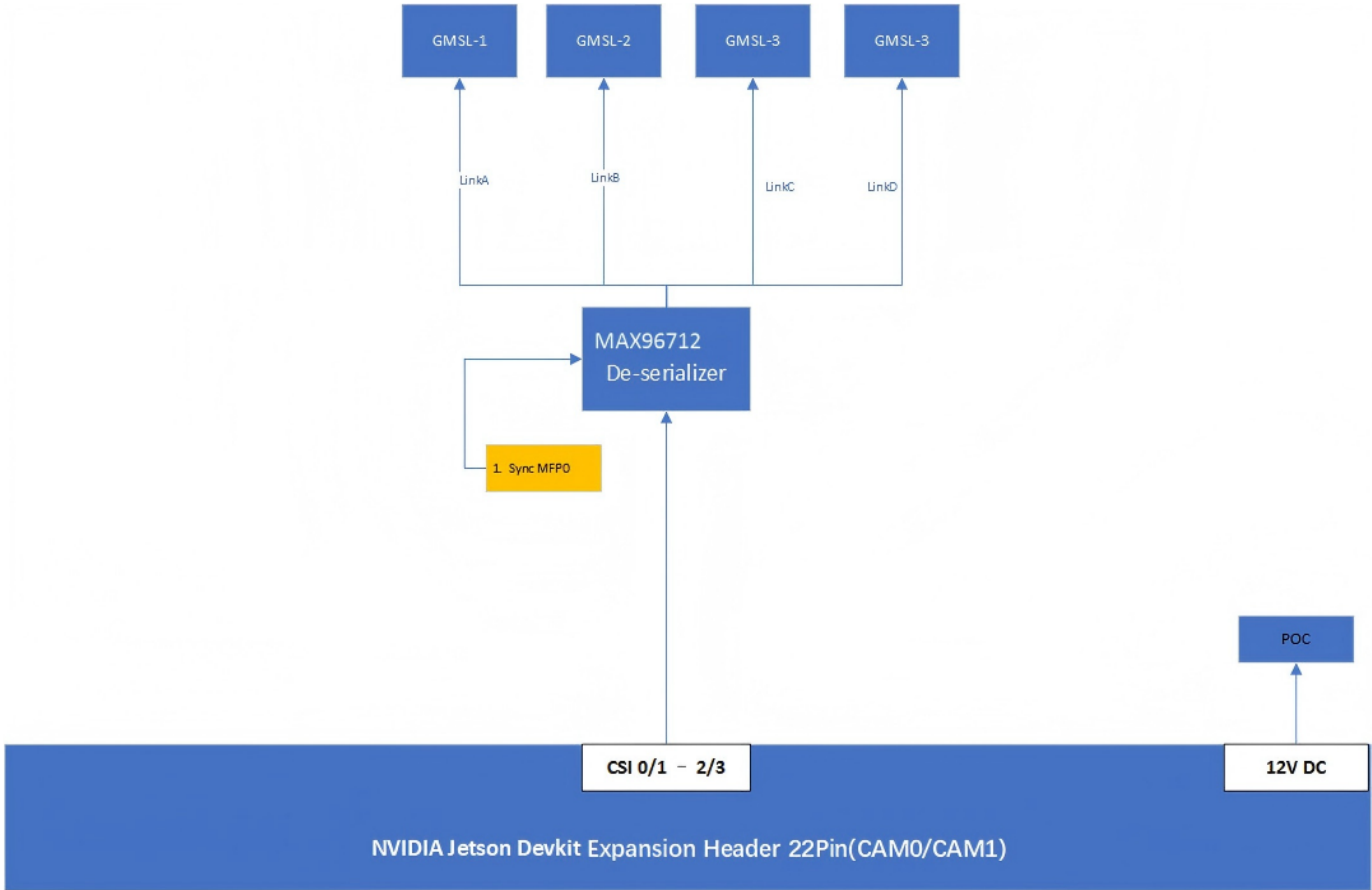
The power supply of the power adapter must be properly grounded. In special scenarios, grounding screws must be installed for direct earth grounding.

ESD Protection

Electronic components and circuits are highly sensitive to electrostatic discharge (ESD). Although our company has designed ESD protection for the main interfaces of the circuit board during product development, it is impossible to provide full ESD protection for all components and circuits. Therefore, it is recommended to follow ESD safety protection measures when handling any circuit board assemblies. ESD safety protection measures include, but are not limited to, the following:

- 1、Keep the carrier board in an anti-static bag during transportation and storage until the time of installation and deployment;
- 2、Discharge static electricity stored in the human body before touching the board: wear an ESD grounding wrist strap;
- 3、Operate the board only in an ESD-safe area;
- 4、Avoid moving the board in carpeted areas.

Product Block Diagram



Notes:

- The I2C bus number is based on the hardware position (matching the pins of connectors J12 and J13) and may not correspond to that listed in the software.
- The coaxial power supply is shared, but each GMSL line is equipped with its own PoC filter.

Component Location Diagram

External Interfaces

No.	Name	Marking	Remarks
1	Power Input Interface	J1	Supports 12V~24V DC input
2	GMSL Input Interface (FAKRA Coaxial Connector)	J2	For GMSL2 (compatible with GMSL1) cameras
3		J3	
4		J4	
5		J5	
6	External Input Trigger Signal Connector	J11	2.54-2*3P
7	Molex 22Pin ZIF Connector (4-Lane CSI) Interface	J12	CSI0
8		J13	CSI1

Interface Functions

Function: <u>POWER Input</u>	Pin	Signal
Marking : <u>J1</u>	1	12V~24V
Type/Model : <u>JPD1030-N521-4F</u>	2	GND
	3	GND

Function: <u>External Input Trigger Signal Connector</u> Marking : <u>J11</u> Type/Model : <u>2.54-2*3P</u>				
Pin	Signal	Pin	Signal	
1	MAX1_ALL_SYNC	4	MAX1_CSI2_SYNC	
2	MAX1_CSI0_SYNC	5	MAX1_CSI3_SYNC	
3	MAX1_CSI1_SYNC	6	GND	

Function: <u>FAKRA Connector</u>				
Marking : <u>J2、 J3、 J4、 J5</u>				
Type/Model: <u>Amphenol Horizontal Single-Ended Coaxial Connector FK1252ZW-031-TLCP5G-50</u>				
Function Overview : GMSL camera application interface is based on V4L2 (Video for Linux, a driver framework for Linux video capture-related devices), which provides a set of unified interface specifications for drivers and applications to facilitate application development. If 4 standard YUV cameras are connected, the corresponding device mapping relationship will be displayed after firmware upgrade as follows.				
Physical Interface	Device Node	Remarks		
GMSL0~3	/dev/video0	Corresponding to IIC-9		
	/dev/video1			
	/dev/video2			
	/dev/video3			
GMSL4~7	/dev/video4	Corresponding to IIC-10		
	/dev/video5			
	/dev/video6			
	/dev/video7			
Marking	Pin	Signal	Pin	Signal
J2	1	MAX1_SIOA_P	2、 3、 4、 5	GND
J3		MAX1_SIOB_P		
J4		MAX1_SIOC_P		
J5		MAX1_SIOD_P		

Function : <u>Molex 22Pin CSI Output</u> Marking : <u>J12 、 J13</u> Type/Model : <u>AFC01-S22FCC-00</u>									
Marking	Pin	Signal	Pin	Signal	Marking	Pin	Signal	Pin	Signal
J12	1	GND	13	GND	J13	1	GND	13	GND
	2	MAX1_CSI0_DATA0_N	14	MAX1_CSI1_DATA1_N		2	MAX1_CSI2_DATA0_N	14	MAX1_CSI3_DATA1_N
	3	MAX1_CSI0_DATA0_P	15	MAX1_CSI1_DATA1_P		3	MAX1_CSI2_DATA0_P	15	MAX1_CSI3_DATA1_P
	4	GND	16	GND		4	GND	16	GND
	5	MAX1_CSI0_DATA1_N	17	MAX1_ALL_SYNC_C		5	MAX1_CSI2_DATA1_N	17	MAX1_ALL_SYNC_C
	6	MAX1_CSI0_DATA1_P	18	NC		6	MAX1_CSI2_DATA1_P	18	NC
	7	GND	19	GND		7	GND	19	GND
	8	MAX1_CSI0_CKA_N	20	MAX1_SCL_CON		8	MAX1_CSI2_CKA_N	20	MAX1_SCL_CON
	9	MAX1_CSI0_CKA_P	21	MAX1_SDA_CON		9	MAX1_CSI2_CKA_P	21	MAX1_SDA_CON
	10	GND	22	VCC_3V3		10	GND	22	VCC_3V3
	11	MAX1_CSI1_DATA0_N	23	GND		11	MAX1_CSI3_DATA0_N	23	GND
	12	MAX1_CSI1_DATA0_P	24	GND		12	MAX1_CSI3_DATA0_P	24	GND

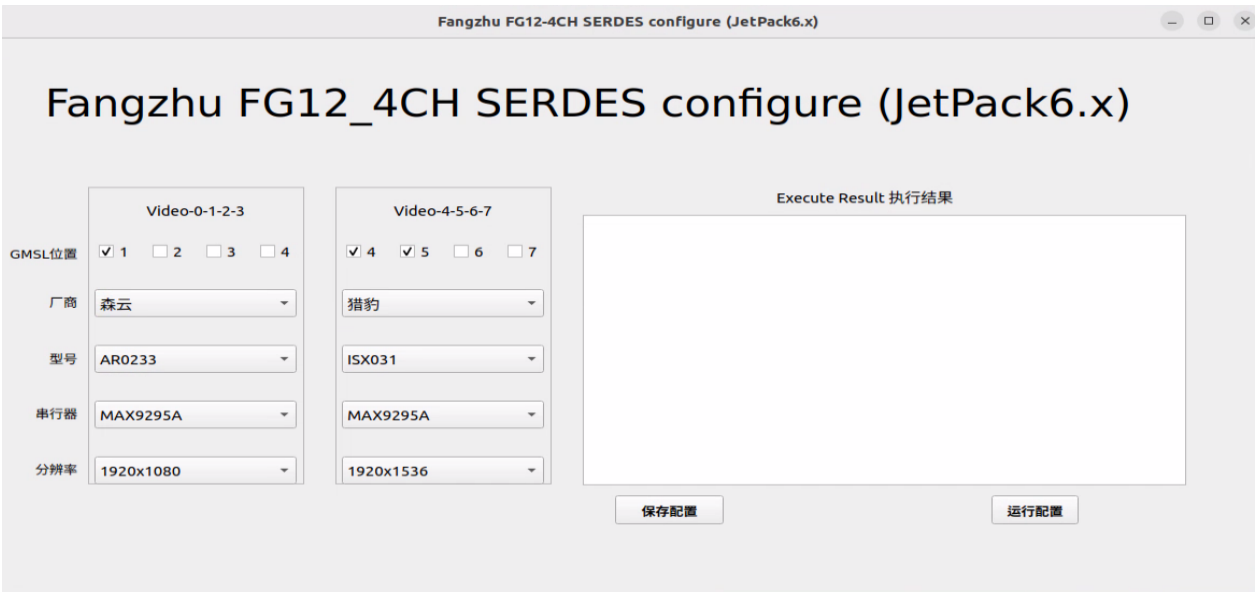
Camera Testing

When the FG12-4CH capture board and cameras are connected to the 4 CAM interfaces corresponding to the I2C-9 bus, select the CAM0~3 configuration.
When the FG12-4CH capture board and cameras are connected to the 4 CAM interfaces corresponding to the I2C-10 bus, select the CAM4~7 configuration.

The capture board supports dynamic connection and loading of cameras. Configure CAM0~7 by running the command `sudo fzcaml_ui`, then select different camera brands and Sensor models.

Operation Steps:

Select the camera node corresponding to the GMSL Position -> Select Manufacturer (厂商)-> Select Model (型号) -> Save Configuration (保存配置) -> Run Configuration (运行配置)



RGB Camera Quick Verification and Reference Code

The device supports video stream output via Gstreamer. The methods for image acquisition and display are as follows:

8M Camera : (3840x2160)

```
gst-launch-1.0 v4l2src device=/dev/video0 ! "video/x-raw, format=(string)UYVY, width=(int)3840, height=(int)2160" ! videoconvert ! fpsdisplaysink video-sink=xvimagesink sync=false
```

5M Camera : (2880x1860)

```
gst-launch-1.0 v4l2src device=/dev/video0 ! "video/x-raw, format=(string)UYVY, width=(int)2880, height=(int)1860" ! videoconvert ! fpsdisplaysink video-sink=xvimagesink sync=false
```

2M Camera : (1920x1080)

```
gst-launch-1.0 -ev v4l2src device=/dev/video0 ! "video/x-raw, format=(string)UYVY, width=(int)1920, height=(int)1080" ! fpsdisplaysink text-overlay=0 video-sink=fakesink sync=0
```

1M Camera : (1280x720)

```
gst-launch-1.0 -ev v4l2src device=/dev/video0 ! "video/x-raw, format=(string)UYVY, width=(int)1280, height=(int)720" ! fpsdisplaysink text-overlay=0 video-sink=fakesink sync=0
```

For cameras with other resolutions, configure the parameters according to the specific camera specifications.

The camera support list can be obtained from the official website or our sales staff.

Warranty Terms of Zhuying Intelligent Robot (Shenzhen) Co., Ltd.

Important Notice

Zhuying Intelligent Robot (Shenzhen) Co., Ltd. warrants that all products supplied are free from defects in materials and workmanship to the best of its knowledge, and fully comply with the specifications of the original factory's official delivery.

The warranty scope of Zhuying Intelligent Robot (Shenzhen) Co., Ltd. covers all original factory products. For malfunctions of accessories configured by distributors, please negotiate a solution directly with the relevant distributor.

Warranty Period: All products provided by the company are covered by a 1-year warranty (lifetime repair services are provided for products beyond the warranty period). The warranty period commences on the date of factory shipment. For products repaired within the warranty period, the warranty for the repaired parts is extended by an additional 12 months. Unless otherwise notified by Zhuying Intelligent Robot (Shenzhen) Co., Ltd., the date on the original factory delivery note shall be deemed as the date of factory shipment.

How to Obtain Warranty Services

If your product fails to operate normally within the warranty period, please contact Zhuying Intelligent Robot Co., Ltd. to obtain warranty services. A valid purchase invoice must be presented when applying for product warranty (this is the proof of your right to obtain warranty services).

Warranty Resolution Measures

When requesting warranty services, you must follow the problem determination and resolution procedures stipulated by Zhuying Intelligent Robot Co., Ltd. You are required to accept the initial diagnosis by our technical staff via phone call, WeChat or email. You shall cooperate to fill in all information on the repair application form provided by us in detail to ensure accurate judgment of the fault cause and damaged parts (a quotation will be provided for out-of-warranty products for your confirmation).

Zhuying Intelligent Robot Co., Ltd. reserves the right to repair or replace the defective product. In case of replacement or repair, the defective product or the defective parts removed during repair/replacement shall be returned to Zhuying Intelligent Robot (Shenzhen) Co., Ltd.

Since some repaired products need to be sent to the original factory, we recommend you purchase transportation insurance to avoid accidental losses. The company shall not be liable for any damage or loss of the shipped items during transportation if the user waives transportation insurance.

For products within the warranty period:

The user shall bear the freight for shipping the defective product to the manufacturer;

Zhuying Intelligent Robot (Shenzhen) Co., Ltd. shall bear the freight for shipping the repaired/replaced product back to the user.

Exclusions from Warranty

The warranty shall not apply to the following situations:

- 1、Malfunctions caused by improper installation, improper use, misuse or abuse of the product (e.g., exceeding the rated working load);
- 2、Product failure or damage caused by improper maintenance/storage (e.g., fire, explosion) or natural disasters (e.g., lightning strike, earthquake, typhoon);
- 3、Modifications to the product (e.g., changes to circuit characteristics, mechanical characteristics, software characteristics, anti-corrosion/anti-moisture/anti-dust treatment);
- 4、Other malfunctions obviously caused by improper use (e.g., overvoltage, undervoltage, reverse polarity connection, bent/broken pins, incorrect bus connection, component detachment, ESD damage, external extrusion, drop damage, overhigh temperature, overhigh humidity, poor transportation);
- 5、Alteration or removal of the product's identification marks and part numbers;
- 6、The product has exceeded the warranty period.

Special Note

If multiple products experience the same malfunction or the same malfunction/damage occurs repeatedly on the same equipment, the company reserves the right to request the user to provide physical objects or technical documents of the peripheral equipment (e.g., monitors, external devices, cables, power supplies, connection diagrams, system structure diagrams) for fault cause analysis and liability confirmation. Otherwise, the company reserves the right to refuse to provide warranty services, and repair services will be charged at the market price with a repair deposit required.